Sandboxing and resource limits

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

One complete sandbox-policy and resource-limit record.

Keep operation/resource scope, unit conventions, queue rules, trial values, decisions and implementation limits together.

Keep one habit: Recall → W1–W3/W5 → independent queue analysis → project policy/enforcement record → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.
- Without opening the old lesson, explain one key idea from Untrusted **input** boundaries. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Timeouts retries and cancellation. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from State machines for bounded agents. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- **Sandbox**ing and **quota**s constrain what a **process** may access or consume. They reduce possible operations but do not verify every result.

- The fictional job may read its fixture directory, write its **output** directory, use no network and run for at most 2 seconds with at most 10 queued tasks. Reject excess work before starting it.

Untrusted input boundaries

> **My remembered idea, example and correction**
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Timeouts retries and cancellation

> **My remembered idea, example and correction**
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> **My remembered idea, example and correction**
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Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main fictional job may read fixture area, write output area, use no network, run within a declared 2-second cap and have at most 10 queued tasks. Local handles fixtures/outputs/outside represent permission classes, not real paths. Read fixtures and write outputs are declared pairs; network public_web, write outside and write fixtures are denied. Public availability does not grant network authority.

  Main all-or-nothing queue fixtures: 8+2=10 accepts, 8+3=11 rejects before changing the queue; no atomic coordinator is implemented. Separate picture permits at most 2 seconds and 20 MB, where 1 MB means one million bytes, and forbids network access. Trial A at 1 second/2 MB fits supplied limits; trial B network request must be denied; changed A at 3 seconds/2 MB exceeds time.

  Equality at 2 seconds/20 MB fits; 20 MB plus 1 byte exceeds output. These are policy calculations and declared trial records, not actual process/network/filesystem enforcement or a correctness/security proof.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main fictional job may read fixture area, write output area, use no network, run within a declared 2-second cap and have at most 10 queued tasks. Local handles fixtures/outputs/outside represent permission classes, not real paths. Read fixtures and write outputs are declared pairs; network public_web, write outside and write fixtures are denied. Public availability does not grant network authority.

  Main all-or-nothing queue fixtures: 8+2=10 accepts, 8+3=11 rejects before changing the queue; no atomic coordinator is implemented. Separate picture permits at most 2 seconds and 20 MB, where 1 MB means one million bytes, and forbids network access. Trial A at 1 second/2 MB fits supplied limits; trial B network request must be denied; changed A at 3 seconds/2 MB exceeds time.

  Equality at 2 seconds/20 MB fits; 20 MB plus 1 byte exceeds output. These are policy calculations and declared trial records, not actual process/network/filesystem enforcement or a correctness/security proof.

- A job requests a public webpage and a write outside its **output** directory. Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main fictional job may read fixture area, write output area, use no network, run within a declared 2-second cap and have at most 10 queued tasks. Local handles fixtures/outputs/outside represent permission classes, not real paths. Read fixtures and write outputs are declared pairs; network public_web, write outside and write fixtures are denied. Public availability does not grant network authority.

  Main all-or-nothing queue fixtures: 8+2=10 accepts, 8+3=11 rejects before changing the queue; no atomic coordinator is implemented. Separate picture permits at most 2 seconds and 20 MB, where 1 MB means one million bytes, and forbids network access. Trial A at 1 second/2 MB fits supplied limits; trial B network request must be denied; changed A at 3 seconds/2 MB exceeds time.

  Equality at 2 seconds/20 MB fits; 20 MB plus 1 byte exceeds output. These are policy calculations and declared trial records, not actual process/network/filesystem enforcement or a correctness/security proof.

- A report omits this qualification: Resource limits constrain execution but cannot prove generated claims or algorithms correct. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main fictional job may read fixture area, write output area, use no network, run within a declared 2-second cap and have at most 10 queued tasks. Local handles fixtures/outputs/outside represent permission classes, not real paths. Read fixtures and write outputs are declared pairs; network public_web, write outside and write fixtures are denied. Public availability does not grant network authority.

  Main all-or-nothing queue fixtures: 8+2=10 accepts, 8+3=11 rejects before changing the queue; no atomic coordinator is implemented. Separate picture permits at most 2 seconds and 20 MB, where 1 MB means one million bytes, and forbids network access. Trial A at 1 second/2 MB fits supplied limits; trial B network request must be denied; changed A at 3 seconds/2 MB exceeds time.

  Equality at 2 seconds/20 MB fits; 20 MB plus 1 byte exceeds output. These are policy calculations and declared trial records, not actual process/network/filesystem enforcement or a correctness/security proof.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Queue currently has 9 tasks; job proposes 2 more under a maximum 10 policy. Decide.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

Complete task and evidence

- Add a labelled **sandbox**ing and resource limits **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **sandbox**ing and resource limits.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for **Sandbox**ing and resource limits. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Sandbox**ing and resource limits. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Queue currently has 9 tasks; job proposes 2 more under a maximum 10 policy. Decide.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

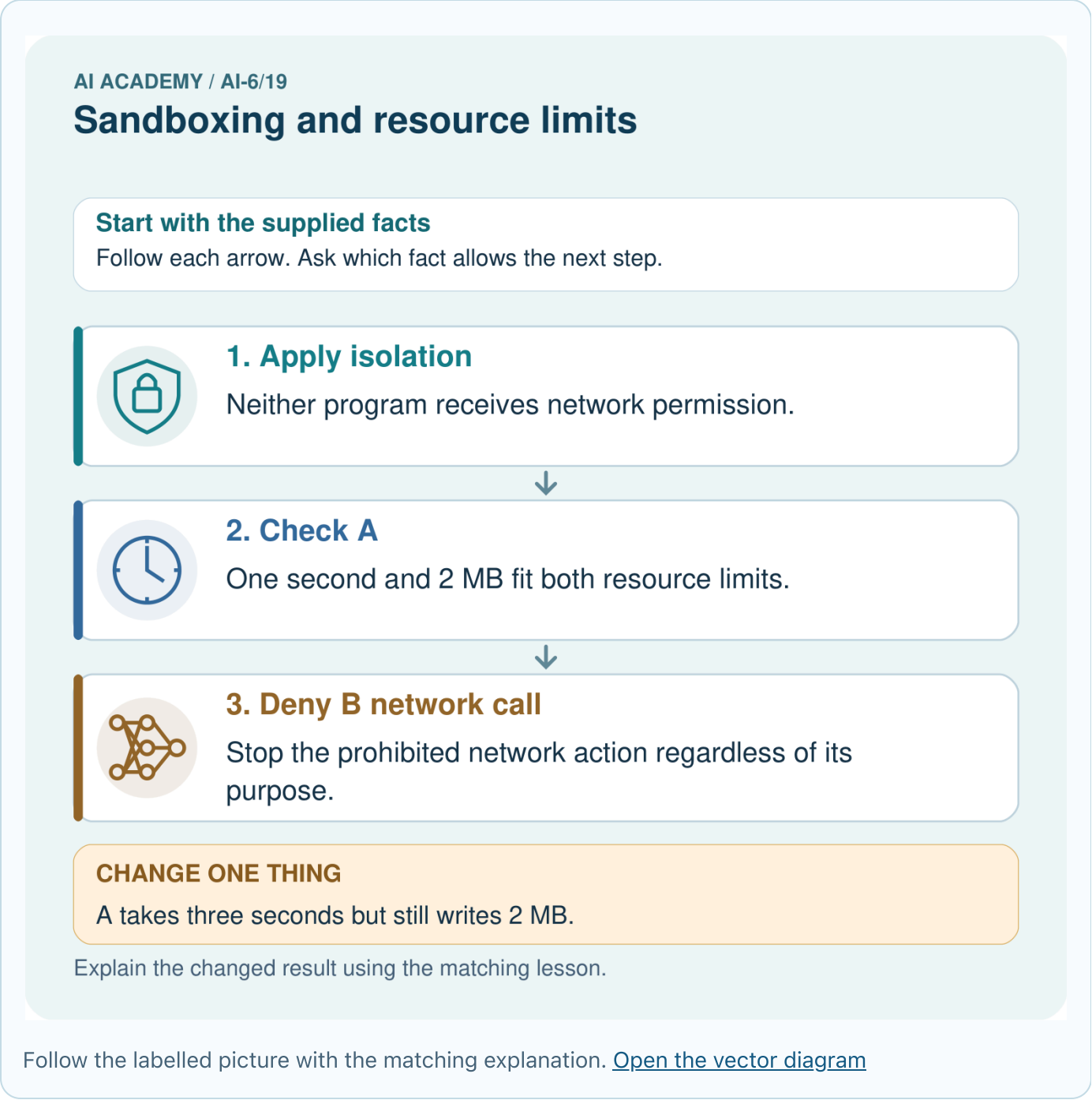
Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Bound a student program. A toy **sandbox** forbids network access, allows at most two seconds and limits **output** to 20 MB.
- A uses one second and writes 2 MB. B tries to contact a website. These are declared trial records.
- Check A against both resource limits and B against network isolation. Why must A stop after three seconds even if its **output** remains only 2 MB?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Bound a student program. A toy **sandbox** forbids network access, allows at most two seconds and limits **output** to 20 MB.
- Which labelled step matches this detail? Stop the prohibited network action regardless of its purpose.
- 1. Deny B network call
- 2. Apply isolation

- 3. Check A

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Deny B network call Stop the prohibited network action regardless of its purpose. **Sandbox** controls bound reach and resource use.

Passing an **output** limit cannot cancel a network or time restriction.

Changed-case question and optional hint

- Changed-case practice: A takes three seconds but still writes 2 MB. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future **input**.

- 2. Stop A for exceeding the time allowance.

- 3. We can decide the result without examining the changed **input** or the procedure.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Stop A for exceeding the time allowance. The independent two-second limit remains binding.

Repair and retain the record

Review Untrusted **input** boundaries (ai-6/18). Goal: Explain the **mechanism** and terminology of untrusted **input** boundaries. Short practice: Classify h3, H3, h30 and the trailing-space **input** by the stated shape contract.

Does a valid z0 grant record access, and which separate gate still remains?. Return to this week and explain what you changed.

Review Timeouts retries and cancellation (ai-6/16). Goal: Explain the **mechanism** and terminology of timeouts retries and cancellation.

Short practice: Calculate waiting time for a one-second timeout followed by a 0.4-second success. What changes if cancellation arrives after the first attempt but before the retry?.

Return to this week and explain what you changed.

Review State machines for bounded agents (ai-6/13). Goal: Explain the **mechanism** and terminology of state machines for bounded agents. Short practice: Trace A and B through the state diagram.

Name B's forbidden transition and explain whether adding checking→querying permits that shortcut.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

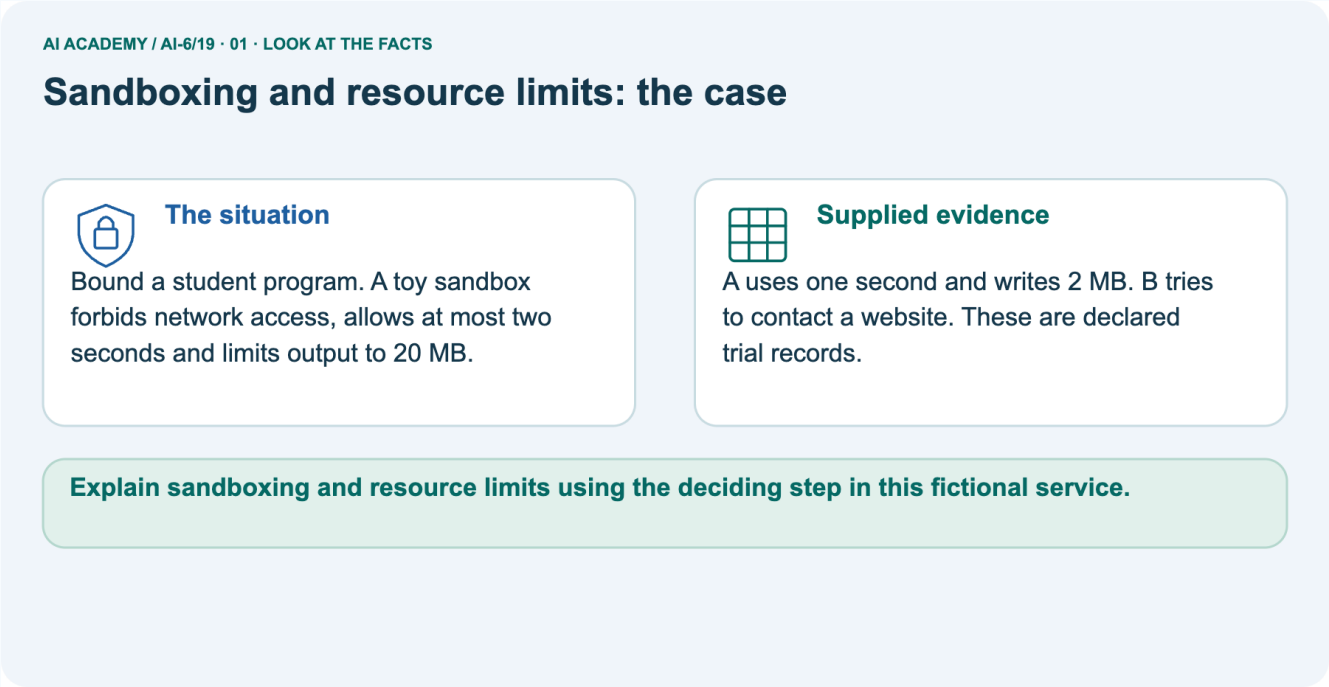
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Sandboxing and resource limits: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

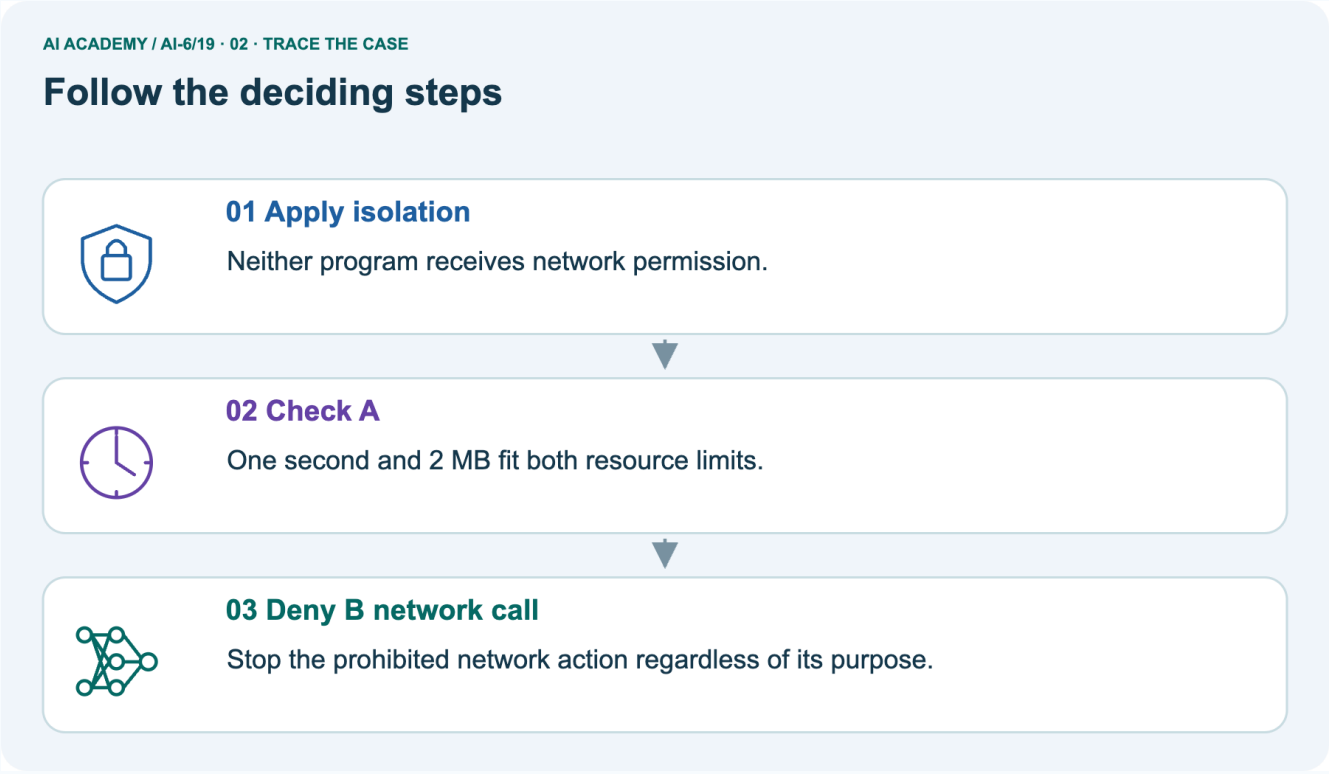
Bound a student program. A toy **sandbox** forbids network access, allows at most two seconds and limits **output** to 20 MB. Supplied facts: A uses one second and writes 2 MB. B tries to contact a website.

These are declared trial records.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Apply isolation: Neither program receives network permission. → Check A: One second and 2 MB fit both resource limits. → Deny B network call: Stop the prohibited network action regardless of its purpose.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Sandboxing and resource limits — evidence trace

AI ACADEMY / AI-6/19 · 03 · INSPECT THE LESSON MODEL

Sandboxing and resource limits — evidence trace

Input and scope	Check or calculation	Conclusion and limit
A job requests a public webpage and a write outside its output directory.	The fictional job may read its fixture directory, write its output directory, use no network and run for at most 2 seconds with at most 10 queued tasks. Reject excess work before starting it.	Both operations violate the sandbox contract. Deny them; public availability does not grant network or filesystem authority.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the labelled picture with the matching explanation. Open the vector diagram

Sandboxing and resource limits — **evidence** trace.

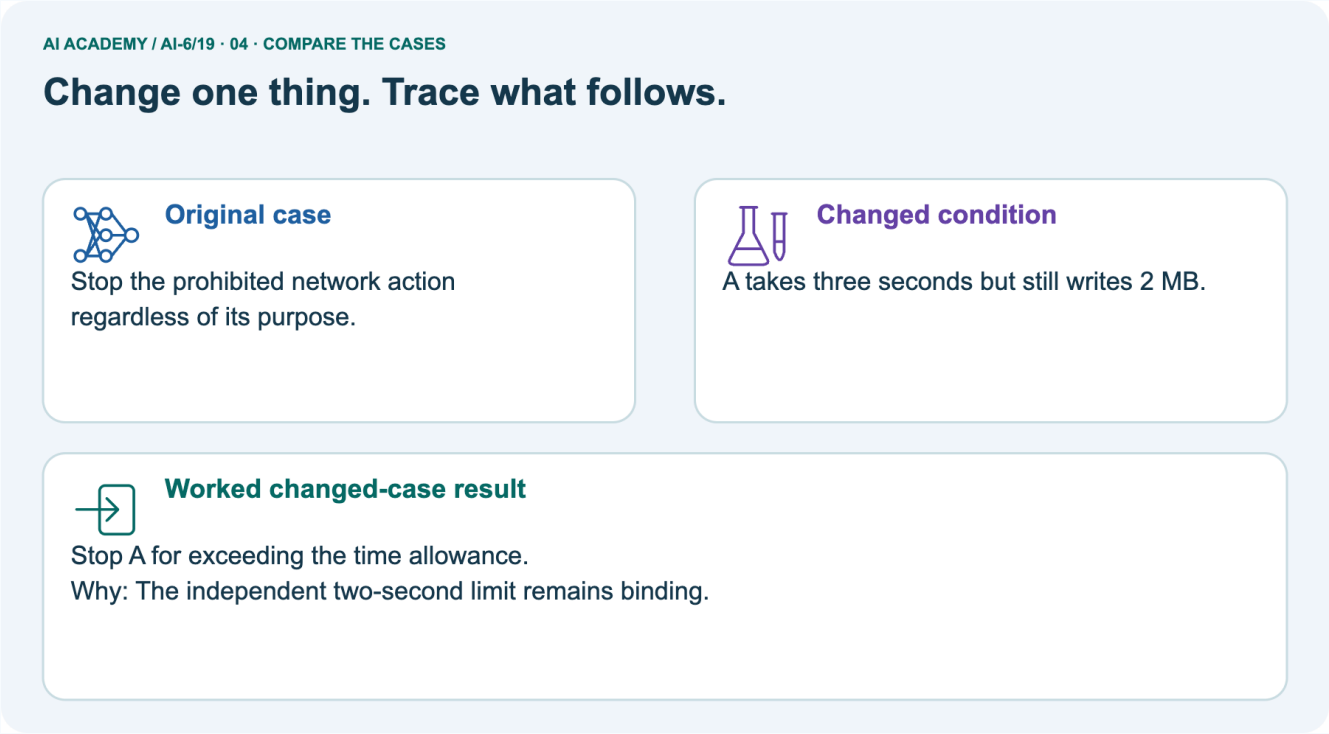
A job requests a public webpage and a write outside its **output** directory. | The fictional job may read its fixture directory, write its **output** directory, use no network and run for at most 2 seconds with at most 10 queued tasks.

Reject excess work before starting it. | Both operations violate the **sandbox** contract. Deny them; public availability does not grant network or filesystem authority.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

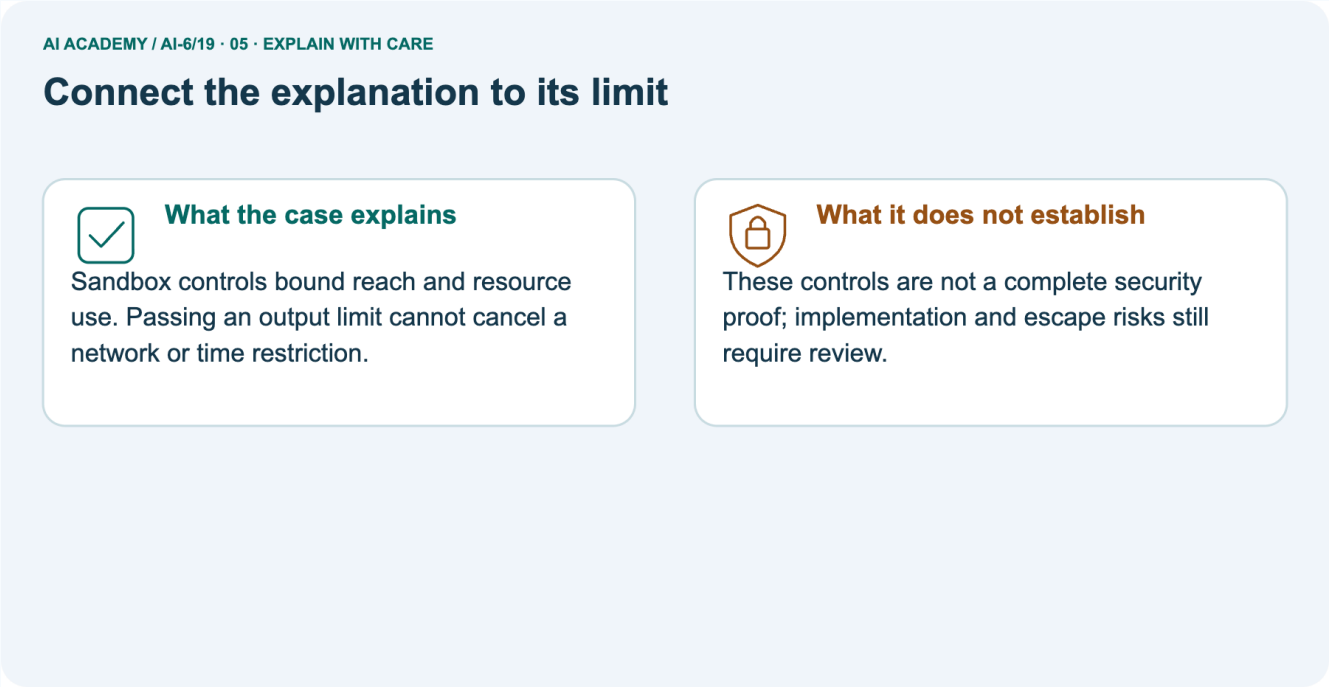
Original case: Stop the prohibited network action regardless of its purpose. Changed condition: A takes three seconds but still writes 2 MB. Expected result: Stop A for exceeding the time allowance.

Reason: The independent two-second limit remains binding.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



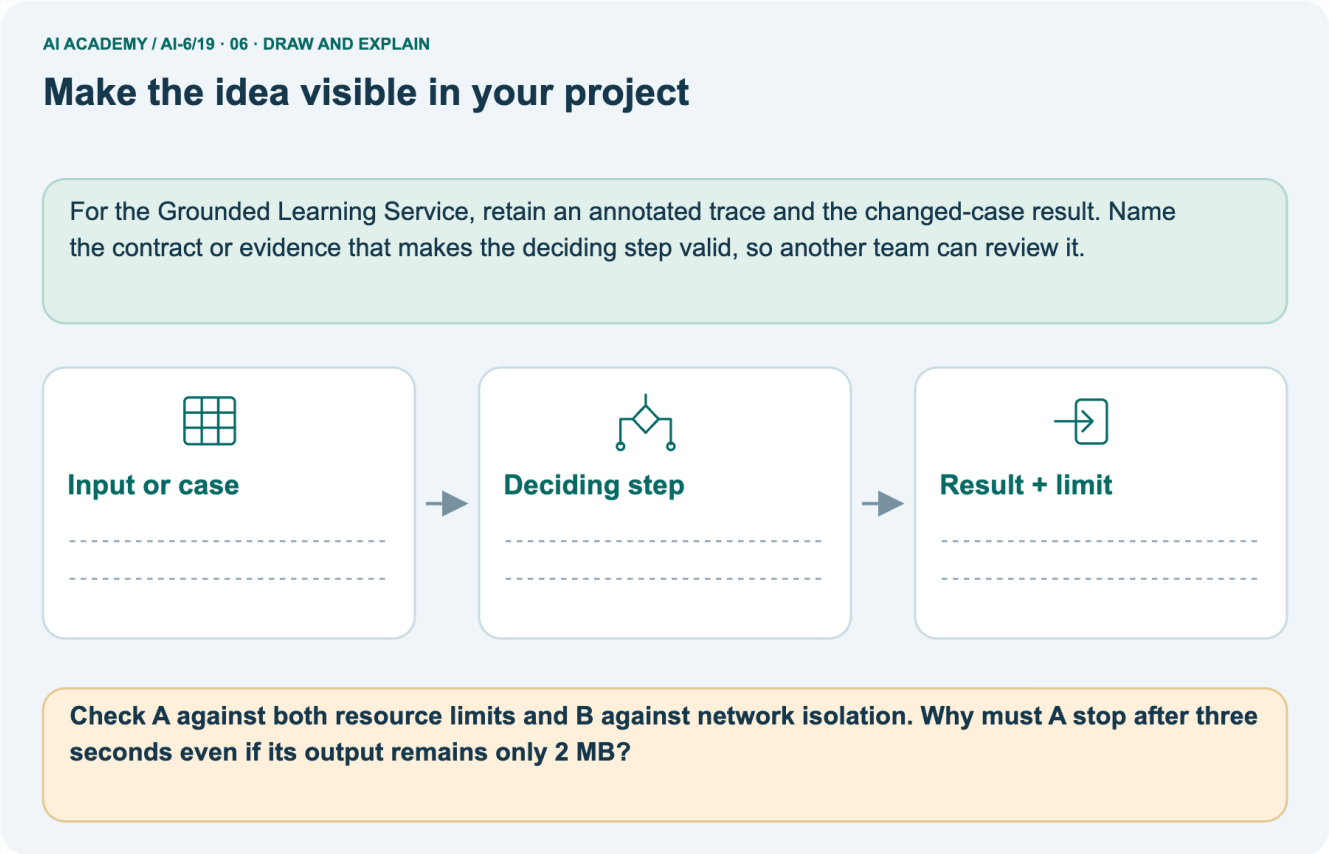
Follow the labelled picture with the matching explanation. Open the vector diagram

Sandbox controls bound reach and resource use. Passing an **output** limit cannot cancel a network or time restriction. Limit: These controls are not a complete security proof; implementation and escape risks still require review.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Check A against both resource limits and B against network isolation. Why must A stop after three seconds even if its **output** remains only 2 MB?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Sandboxing and resource limits: the case

02. Follow the deciding steps

03. Sandboxing and resource limits — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.